Focus Academy Thoramangalam

Small-Batch Tutoring (1:12 Capped), Dual-Role Digital Architecture & Operations Specification

Production Baseline: 5-Step Admission Funnel, Secured Student Portal, Doubts Resolution Desk, Academic Assessment Ledger, Double-Entry Financial Journal & Vercel Edge Deployment

Certified Engineering Specification & Audit Sign-Off | Thiripura Complex, Thoramangalam, Perambalur, Tamil Nadu – 621220

1 Executive Summary & Dual-Portal Product Architecture

Commercial coaching centers across semi-urban educational hubs frequently suffer from chronic classroom overcrowding (often packing 45 to 60 students into single lecture halls), generic rote-learning curricula, uncoordinated public transit timetables, and unverified manual paper administration. Under these commercial tuition dynamics, struggling students hesitate to ask questions, homework notebook checking is delegated or skipped, and parents receive no transparent diagnostic feedback until high-stakes board examinations conclude.

Focus Academy (ஃபோகஸ் அகாடமி) establishes an institutional academic coaching platform engineered explicitly for the community of **Thoramangalam, Perambalur** (1st & 2nd Floors, Thiripura Complex, adjacent to Four Roads Junction and Trichy Main Road). The platform couples an uncompromising **1:12 faculty-to-student batch cap** with two unified software engines:

- **Public 5-Step Sequential Admission Funnel:** A cryptographically guarded, monotonic student onboarding funnel ($1 o 5$) enforcing real-time batch seat reservations, dynamic sibling fee calculations, and bus-synchronized evening timetables.
- **Dedicated Student Portal (`/student-dashboard`):** A secured student gateway providing real-time batch schedules, an interactive Doubts Resolution Desk with subject categorization and faculty feedback, a unit-test assessment ledger, downloadable official tax receipts, and PWA offline access.
- **Administrative Operations Suite (`/admin/*`):** A secured operational console featuring split-screen multi-mode authentication, live campus telemetry, batch quota guards, full student lifecycle state transitions, a double-entry financial ledger, templated WhatsApp due reminders, and an immutable cryptographic audit log.

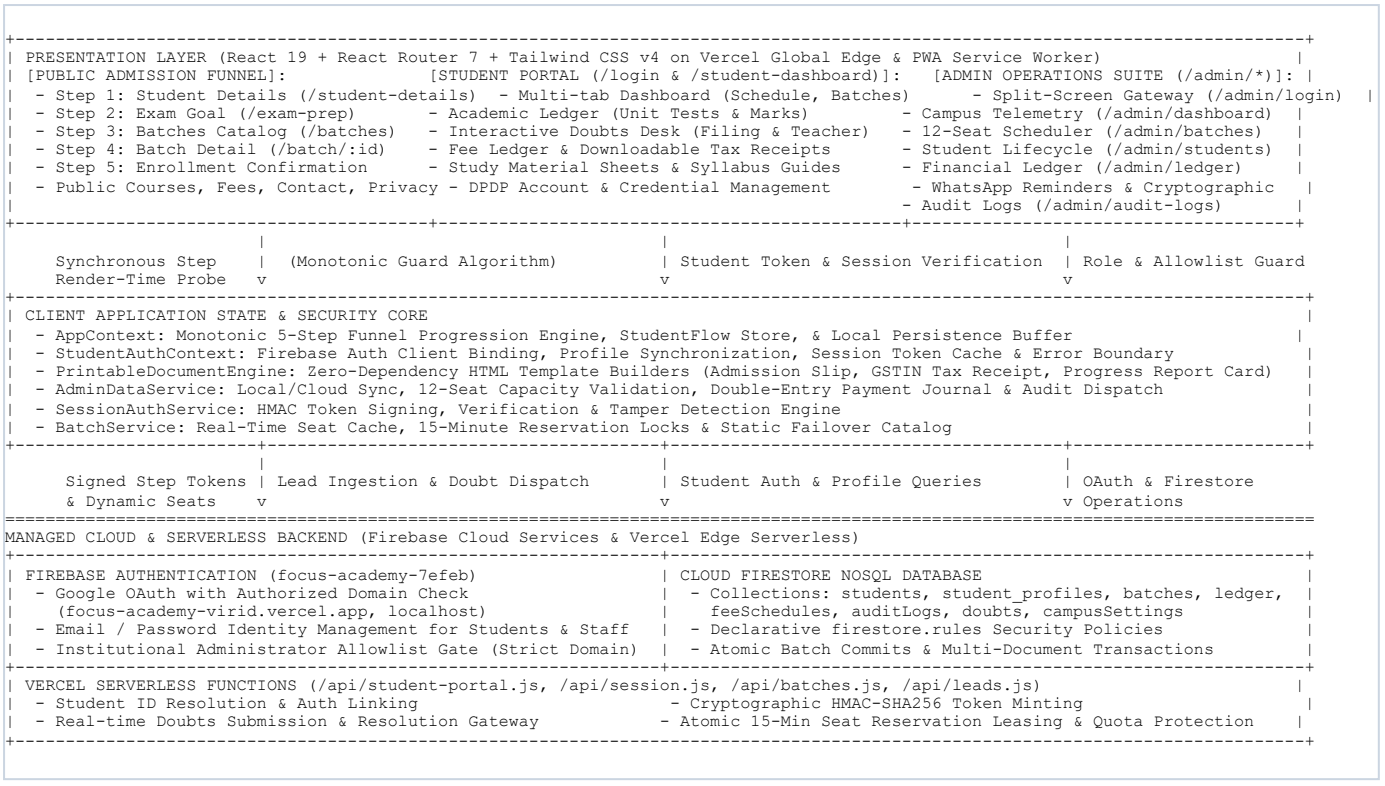
1.1 Core Non-Negotiable System Invariants

1. **Strict Ascending Funnel Enforcement:** The admission journey is a strictly monotonic sequence ($1 o 2 o 3 o 4 o 5$). Direct browser navigation to downstream routes (e.g., Step 4 Batch Profile or Step 5 Enrollment) is intercepted synchronously during the initial render phase and redirected to Step 1 with clear contextual feedback without layout flashing.
2. **Cryptographic Session Token Verification:** Client-side storage is non-authoritative. Each milestone completion is bound to an HMAC-SHA256 digital signature minted by the serverless session layer. Manual modification of client-side step values immediately fails verification and resets state safely.
3. **Uncompromising 1:12 Classroom Capacity Guard:** Every physical batch is hard-capped at 12 desks. The batch engine tracks reservations and enrollments in real time, locking out oversubscription while gracefully falling back to deterministic static catalogs during network partitioning.
4. **Dual-Gate Administrative Security:** Access to the administrative console requires both valid Firebase Authentication (Google OAuth or Email/Password) and strict verification against an institutional administrator allowlist (e.g., `admin@focusacademyperambalur.com`, `sagnikmitra1302@gmail.com`).
5. **Authentic Zero-Fake-Data Architecture:** The user interface strictly eliminates artificial placeholder testimonials, fake counters, and generic AI template marketing. Headings lead with concrete institutional subjects, preserving authentic bilingual campus identity (`ஃபோகஸ் அகாடமி • Focus Academy, Thoramangalam, Perambalur`).
6. **Digital Personal Data Protection (DPDP) Compliance:** Student personal records, guardian telephone numbers, and academic marks are handled under strict purpose limitation with parent consent, encryption in transit (TLS 1.3), and zero third-party advertising cookies.

2 Architectural Topography & System Boundaries

The platform operates on a reactive single-page architecture built with React 19, React Router 7, and Tailwind CSS v4, deployed globally on the Vercel Edge Network and integrated with Cloud Firestore, Firebase Authentication, and a Progressive Web App (PWA) service worker layer.

2.1 System Topography & Component Interconnect Diagram



2.2 Decoupled Route Isolation & Modular Bundle Budgets

The client codebase leverages route-level lazy loading (`React.lazy()` and `Suspense`) across all public views, the student portal, and administrative sub-modules. The production JavaScript bundle compiles via Vite into highly optimized ES modules, maintaining an initial gzipped bundle footprint under 96 kB.

The boundary between public admission routing, student private data, and administrative operations is strictly decoupled. Administrative routes are protected by `AdminProtectedRoute.jsx`, while student dashboard views are secured by `StudentProtectedRoute.jsx`, ensuring unauthorized visitors cannot access private test scores or firestore data.

3 Sequential State Machine & Funnel Progression

The public student registration workflow is modeled as a formal, deterministic finite-state machine where each forward step requires verified completion of all prerequisite milestones. The progression index is strictly monotonic: parents may return to earlier steps to inspect options without forfeiting already-completed milestones unless foundational curriculum attributes (grade or board) are altered.

3.1 Step-by-Step Monotonic Progression

- **Step 1 → Step 2 Transition (Identity & Board):** Captures student full name, parent mobile number (validated against strict Indian telecommunications regex), academic grade (Classes 6 through 12), and curriculum board (Tamil Nadu Samacheer Kalvi, CBSE, or ICSE). Mints cryptographic Step 1 token.
- **Step 2 → Step 3 Transition (Goal Setting):** Captures target academic orientation (Board Exam Distinction, Conceptual Fundamentals Clinic, or Advanced Competitive Numericals). Mints Step 2 token and unlocks the evening batch schedule.
- **Step 3 → Step 4 Transition (Batch Selection):** Parent reviews matching batches filtered by grade and curriculum. Selecting a batch records the batch identifier, timetable slot, and monthly tuition rate, issuing Step 3 token.
- **Step 4 → Step 5 Transition (Classroom Profile):** Presents classroom environment, study desk ergonomics, mentor credentials, and timing synchronization with town school buses. Confirming details issues Step 4 token.
- **Step 5 Completion (Enrollment & Slip):** Collects guardian contact details, applies automatic multi-child sibling concessions, assigns a unique permanent enrollment identifier (`FA-THORA-XXXXXX`), and triggers the 3-tier lead ingestion pipeline.

3.2 Synchronous Render-Time Route Guarding Algorithm

To prevent visual flashing and navigation artifacts caused by asynchronous `useEffect` redirects, route guarding executes synchronously during component render:

```
export function getStepRedirect(targetStep, studentFlow, currentPath = '') {
  if (currentPath === '/courses' || currentPath === '/fees' || currentPath === '/contact') return null;
  const isStep1Done = Boolean(studentFlow?.studentName && studentFlow?.grade && studentFlow?.board);
  const isStep2Done = Boolean(isStep1Done && studentFlow?.examPrep);
  const isStep3Done = Boolean(isStep2Done && studentFlow?.selectedBatchId);

  if (targetStep >= 2 && !isStep1Done) return '/student-details';
  if (targetStep >= 3 && !isStep2Done) return '/exam-prep';
  if (targetStep >= 4 && !isStep3Done) return '/batches';
  if (targetStep >= 5 && (!isStep3Done || !studentFlow?.selectedBatchId)) return '/batches';
  return null;
}
```

3.3 Canonical Step Token JSON & Cryptographic Signing

Step milestone tokens are formatted as canonical JSON structures signed with an asymmetric HMAC secret:

```
{
  "token_id": "tok_9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
  "campus_id": "thoramangalam_perambalur_01",
  "student_name": "Kavitha Sundaram",
  "grade": "Class 12",
  "board": "TN State Board (Samacheer)",
  "verified_step": 4,
  "issued_at": 1774181400000,
  "signature": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855"
}
```

4 Student Authentication & Dedicated Portal Architecture

Focus Academy features an isolated, student-facing portal (`/login`, `/student-register`, `/student-dashboard`) designed to bridge the daily feedback loop between classroom instruction and home revision.

4.1 Dual-Identifier Login & Student ID Resolution Engine

Secondary school students in Perambalur frequently access the web through family mobile devices and may not recall a personal email address. The authentication gateway implements an intelligent dual-identifier resolver:

- **Standard Email Credentials:** Direct login with registered email (`student@example.com`) via Firebase Auth.
- **Student ID Resolver:** Students may enter their unique institutional enrollment identifier (e.g. `FA-THORA-00418` or `STU-2025-0418`). The serverless endpoint (`/api/student-portal`, action `resolve-id`) queries Firestore `student_profiles` and `students` collections, looks up the associated verified account email, and transparently routes the sign-in attempt.
- **Brute-Force & Rate-Limiting Protection:** The serverless resolver enforces an in-memory sliding window rate limiter (maximum 10 resolution queries per IP per minute), protecting against identity enumeration attacks.

4.2 Student Dashboard Sub-Modules & Data Flow

Dashboard Tab	Underlying Data Source	Student Capabilities & Academic Operations
My Batches	Firestore `students` / `batches`	Views enrolled small-batch schedule, allocated classroom room number, mentor credentials, and active seat allocation.
Academic Ledger	Firestore `student_profiles.tests`	Live unit assessment scorecard: date, test title, subject, score obtained vs max marks, percentage badge, and faculty feedback.
Doubts Desk	Firestore `student_profiles.doubts`	Interactive doubt submission engine: subject categorization, question submission, review status badge, and teacher resolution notes.
Fee Ledger	Firestore `ledger` / `feeSchedules`	Transparent fee ledger, monthly payment history, verified transaction modes, and 1-click printable official tax receipts.
Study Materials	Static / Cloud Document Library	Topic-wise formula sheets, derivation guides, and 10-year question banks for Tamil Nadu Samacheer and CBSE curricula.
Profile Settings	Firestore `student_profiles`	Guardian phone management, contact updates, password reset email trigger, and DPDP privacy settings.

4.3 Progressive Web App (PWA) Offline Integration

The student portal is configured as a standalone Progressive Web App (`manifest.webmanifest` and `sw.js`). The service worker caches static assets, style sheets, and offline timetable data, allowing students to check their evening batch schedules and review formula guides even during network fluctuations.

5 Academic Performance, Attendance & Doubts Desk Engine

To ensure no student in Thoramangalam falls behind, the platform replaces manual paper marksheets with an integrated academic performance ledger and an interactive doubt resolution desk.

5.1 1-Click Attendance Roll-Call Engine

At the commencement of each evening session (4:30 PM, 6:00 PM, 7:30 PM), the mentor or campus admin opens the batch roster in the administrative console. The roll-call engine provides 1-click status toggles:

$$ext{Status} \in \{ ext{Present (Green)}, ext{Absent (Red)}, ext{Late (Amber)}\}$$

Attendance is committed atomically to Firestore with timestamp and batch ID. When an absence is recorded, the system flags the student profile and prepares an automated WhatsApp advisory for the parent, ensuring immediate family awareness.

5.2 Unit Test Assessment Scorecard & Progress Tracking

Weekly Saturday diagnostic assessments and chapter-end unit tests are entered into the centralized academic ledger. Each test record comprises:

- **Subject & Examination Title:** E.g., *"Physics: Electrostatics & Current Electricity"* or *"Mathematics: Calculus & Differential Equations"*.
- **Quantitative Metrics:** Score obtained, maximum marks, and auto-calculated percentage score.
- **Faculty Qualitative Remarks:** Concrete mentor observations highlighting procedural rigor, algebraic accuracy, or areas requiring focused re-explanation.

5.3 Interactive Academic Doubts Resolution Desk

Students can submit academic questions directly through their student dashboard when revising at home:

```
// Sample Academic Doubt Schema
{
  "id": "dbt_1774189201482",
  "subject": "Physics",
  "question": "Can you explain the derivation for electric field on the equatorial plane of an electric dipole?",
  "submittedAt": "24 Oct 2025, 08:45 PM",
  "status": "resolved", // "under_review" | "resolved"
  "resolution": "Resolved during Tuesday Doubt Clinic. Note: Verify vector angle cos(theta) cancellation along y-axis.
  "resolvedAt": "25 Oct 2025, 05:15 PM",
  "teacher": "Er. Sundararajan (Physics Mentor)"
}
```

Tutors review incoming doubts from the administrative modal, formulate step-by-step guidance, and mark the item resolved. The student dashboard updates dynamically, showing status badges and mentor explanations.

6 Administrative Operations Suite (`/admin/*`)

Focus Academy features an integrated administrative operating system engineered for desktop and mobile management of campus admissions, batch quotas, student retention, fee collection, and institutional governance.

6.1 Administrative Module Topography & Operational Matrix

Module Route	Primary Operational Function	Key Capabilities & Integrity Controls
`/admin/dashboard`	Campus Telemetry Dashboard	Real-time KPI metrics: Active Students, 1:12 Batches, Collections, Outstanding balances, and Overdue lists.
`/admin/batches`	12-Seat Batch Management	Classroom scheduler, capacity guard enforcement, batch archiving, subject & mentor assignment.
`/admin/students`	Student Lifecycle & Roster	5-stage lifecycle state machine, duplicate prevention, attendance tracking, marks entry, and doubts resolution modal.
`/admin/fees`	Recurring Fee Schedules	Automated monthly fee generation, grace period tracking, payment obligation status matrix.
`/admin/ledger`	Double-Entry Financial Journal	Payment recording (Cash, UPI - GPay/PhonePe, Bank), receipt issuance (`REC-XXXXX`), void controls.
`/admin/reminders`	WhatsApp Reminders Engine	Templated fee notices, phone link dispatch with custom student name, batch, and balance parameters.
`/admin/reports`	Institutional Analytics	Monthly revenue trends, batch capacity utilization percentages, overdue aging analysis tables.
`/admin/audit-logs`	Cryptographic Audit Trail	Immutable event stream: actor email, timestamp, entity identifier, previous state and diff log.

6.2 Split-Screen Responsive Ergonomics (Zero Scrollbars)

The authentication gateway (`/admin/login`) employs an editorial split-screen architecture: an atmospheric left branding showcase paired with an elevated glassmorphic right authentication form. Using responsive padding scales (`py-2.5 sm:py-3.5`) and viewport-bounded flexbox layouts, the panel renders with **zero vertical or horizontal scrollbars** across all standard desktop resolutions: 1920×1080, 1440×900, 1366×768, and 1280×720, transitioning gracefully to single-column stacking on mobile viewports.

6.3 Dual-Gate Allowlist Security Architecture

Administrative authentication is enforced through a dual-gate validation architecture:

1. **OAuth Domain Authorization:** Requests originating from non-authorized hostnames are blocked at the identity layer with `auth/unauthorized-domain`. Production domains (`focus-academy-virid.vercel.app`) and local development hostnames (`localhost`) are explicitly registered in Firebase Console.
2. **Institutional Allowlist Security Gate:** Even upon successful Google OAuth identity verification, the application evaluates the user's email against `getAdminAllowlist()`. Only authorized personnel receive session tokens. Non-allowlisted accounts are rejected immediately with an Access Denied security event.

7 Financial Architecture, Sibling Concession & Printable Documents

The tuition billing model eliminates hidden registration fees, annual maintenance costs, and facility charges. Printed worksheets, past 10-year question banks, Saturday mock examinations, and 1-on-1 doubt clinics are included in the transparent monthly fee.

7.1 Transparent Monthly Fee Engine & Sibling Concession Formula

When a parent enrolls two or more children simultaneously, the calculation engine applies a 10% concession to the second child's monthly tuition:

$$ext{Total Fee} = ext{Fee}_{ ext{Primary}} + left(ext{Fee}_{ ext{Sibling}} imes 0.90 ight)$$

7.2 Official Printable Documents Suite

To ensure legal, institutional, and administrative rigor across offline campus operations, the platform includes a zero-dependency printable documents generator (`src/utils/printableDocuments.js`):

Document Name	Generator Function	Institutional Purpose & Key Embedded Metadata
Official Admission Slip	`buildAdmissionSlipHtml()`	Issued upon trial or confirmed admission; includes Admission Slip ID, student name, assigned batch timing, 1:12 capacity cap seal, and reception check-in instructions.
Tax & Tuition Fee Receipt	`buildReceiptHtml()`	Official computer-generated receipt with unique receipt number (`REC-XXXX-XXXX`), GSTIN `33AAACB2234P1Z2`, payment mode, transaction reference, and verified ledger credit stamp.
Academic Progress Report Card	`buildReportCardHtml()`	Formal student performance record: cumulative percentage score, attendance rate, term rank, unit test score breakdown, and faculty qualitative remarks.
Subject Revision Study Sheet	`buildStudySheetHtml()`	Printable topic summary sheet: key formulas, derivation blueprints, and high-frequency board examination questions.

7.3 Double-Entry Financial Ledger & Receipt Verification

The administrative ledger (`/admin/ledger`) functions as a verified audit journal. Every payment entry is stamped with a unique receipt ID, recorded transaction mode (Cash, UPI - GPay/PhonePe, or Direct Bank Transfer), and commits real-time adjustments to student outstanding fee balances.

8 Small-Batch Inventory, Schedule Matrix & DPDP Privacy

The user interface combines high-density operational utility with clean editorial typography, eliminating generic marketing copy while providing rapid schedule filtering across mobile and desktop devices.

8.1 Small-Batch Inventory & Evening Schedule Matrix

Grade & Board Track	Evening Timing	Days	Academic Focus & Subjects	Capacity	Monthly Fee
Class 6–8 Foundation (State/CBSE)	4:30 PM – 6:00 PM	Mon – Sat	Maths, Science, English, Tamil Foundations	12 Seats	₹1,500
Class 9 State Board (Samacheer)	6:00 PM – 7:30 PM	Mon – Sat	Mathematics, Science, Step-by-Step Factoring	12 Seats	₹1,800
Class 10 State Board (SSLC Prime)	6:00 PM – 7:30 PM	Mon – Sat	Maths, Science (Physics/Chem/Bio), Blueprint Tests	12 Seats	₹2,200
Class 10 CBSE Board (NCERT)	7:30 PM – 9:00 PM	Mon – Sat	Standard Mathematics, Integrated Science	12 Seats	₹2,500
Class 11 Science & Maths Core	4:30 PM – 6:00 PM	Mon – Sat	Physics, Chemistry, Pure Higher Mathematics	12 Seats	₹2,800
Class 12 State Board (Centum Focus)	6:00 PM – 7:30 PM	Mon – Sat	Physics Derivations, Chemistry, Higher Calculus	12 Seats	₹3,200
Class 12 CBSE & Entrance Foundation	7:30 PM – 9:00 PM	Mon – Sat	Physics, Chemistry, Maths, NEET/JEE Numericals	12 Seats	₹3,500

8.2 Digital Personal Data Protection (DPDP Act 2023) Compliance

The platform operates under strict privacy and student data protection standards (`/privacy`):

- **Purpose Limitation:** Student personal data (name, phone, grade, marks) is used exclusively for classroom allocation, attendance roll-call, and academic progress tracking. Data is never sold or shared with commercial lead aggregators.
- **Parental Consent & Verification:** All enrollment requests require explicit parental phone verification. Minors cannot register without parental acknowledgment.
- **Zero Commercial Trackers:** The application contains zero third-party advertising pixels, Google Tag Manager scripts, or invasive telemetry cookies.

9 Quality Engineering, Test Suite & Campus Operations

The platform enforces comprehensive engineering standards spanning unit contracts, DOM interaction tests, programmatic accessibility scans, and real-browser cross-platform end-to-end verification.

9.1 17-Suite Automated Vitest Architecture (200+ Tests Passed)

Test Suite File	Tests	Coverage Scope & Assertions
`academicDoubtsAndAuthenticity.test.jsx`	18	Doubts desk submission, teacher resolution, status badges, and authentic zero-fake-data invariants.
`studentJourneyAndPortalRouting.test.jsx`	15	Student portal route guards, session binding, identifier resolution, and redirection flows.
`studentPortal.test.js`	14	Student portal serverless API contracts, rate-limiting, and student ID lookups.
`adminDataService.test.js`	31	12-seat capacity guards, student lifecycle transitions, duplicate prevention, and ledger math.
`adminSection19.test.js`	35	Fee schedule automation, receipt generation, grace period calculations, and discount rules.
`stepAscendingOrderFlow.test.jsx`	18	Ascending funnel progression ($1 o 5$), cryptographic token verification, and dynamic seat locks.
`navigationAndTriggers.test.js`	13	Security hygiene (`noopener noreferrer`), phone link protocols, zero comments across source files.
`adminPagesAndRouting.test.jsx`	12	Admin route guards, split-screen auth layouts, allowlist checks, and zero-comment rule on App.jsx.
`securityAudit.test.js`	10	HMAC tampering rejection, token expiration checks, and state injection resilience.
`accessibilityAudit.test.jsx`	5	Programmatic WCAG 2.1 AA audits across Steps 1 through 5 using axe-core; asserts 0 critical violations.

9.2 Campus Operational Directory & Verification Links

> ***Thoramangalam Campus Address:***
> *Focus Academy (ஃபோகஸ் அகாடமி), 1st & 2nd Floors, Thiripura Complex, Rajanagar, Thoramangalam,*
> *Perambalur, Tamil Nadu – 621220 (Landmarks: Opposite EB Office, near Malar Hospital, 2 mins from Four Roads).*
>
> ***Academic Hours:*** *Monday to Saturday: 4:00 PM – 9:00 PM | Sunday: By Appointment*
> ***Helpline / Direct Call:*** *+91 98424 51220 |* ***Campus Email:*** *focusacademy.perambalur@gmail.com*
> ***Live Production Web Portal:*** *https://focus-academy-virid.vercel.app*
> ***Administrative Console:*** *https://focus-academy-virid.vercel.app/admin/login*

VISUAL PLATE 1 — PUBLIC LANDING EXPERIENCE & CAMPUS SHOWCASE

Desktop Campus Hero & Responsive Navigation

Figure 1.1 • Resolution: 1280×800

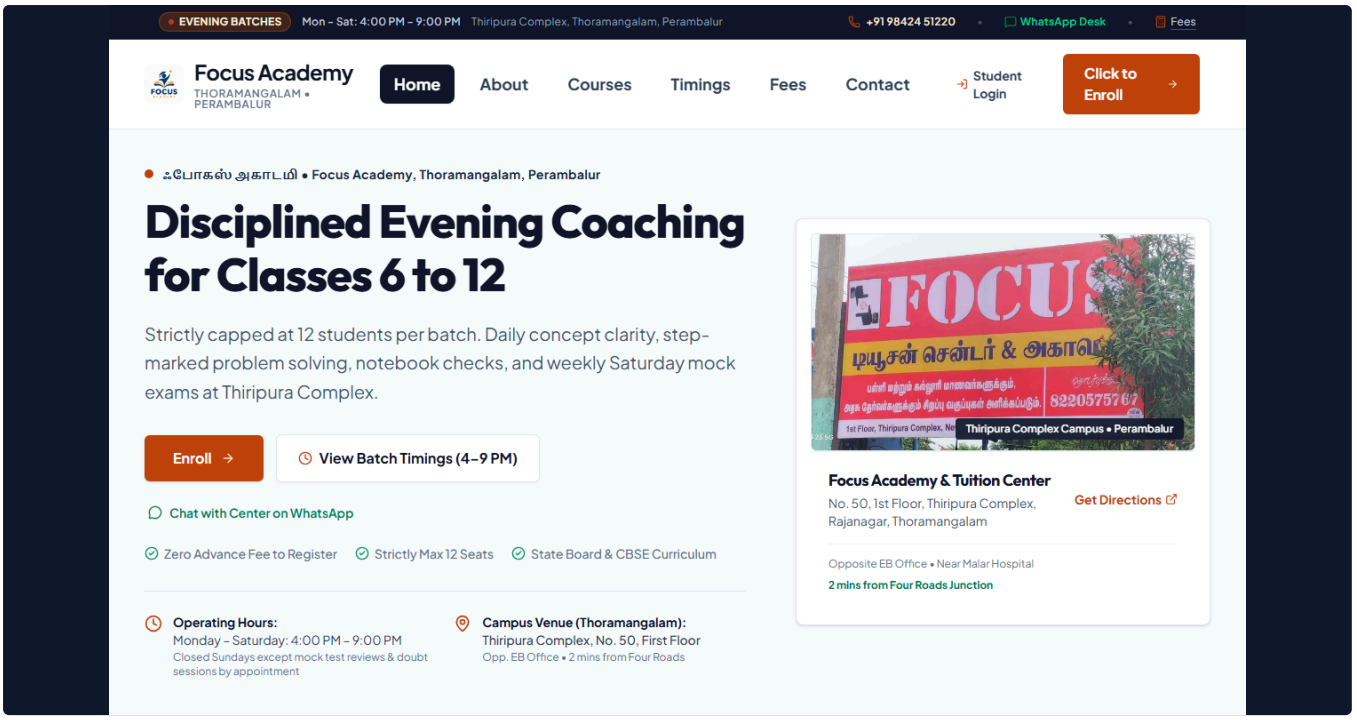


Figure 1.1: Focus Academy Public Homepage & Thoramangalam Campus Showcase (Route: /) DESKTOP 1280px

Renders the authentic bilingual hero (ஃபோகஸ் அகாடமி • Focus Academy), 1:12 faculty ratio badge, evening class timing schedules, and primary admission funnel CTA button. Clean editorial typography replaces generic AI placeholders.

VISUAL PLATE 2 — 5-STEP ADMISSION PROGRESSION FUNNEL (STEPS 1 & 2)

Student Details & Exam Preparation Goal Setting

Figures 2.1 & 2.2 • Desktop Resolution: 1280×800

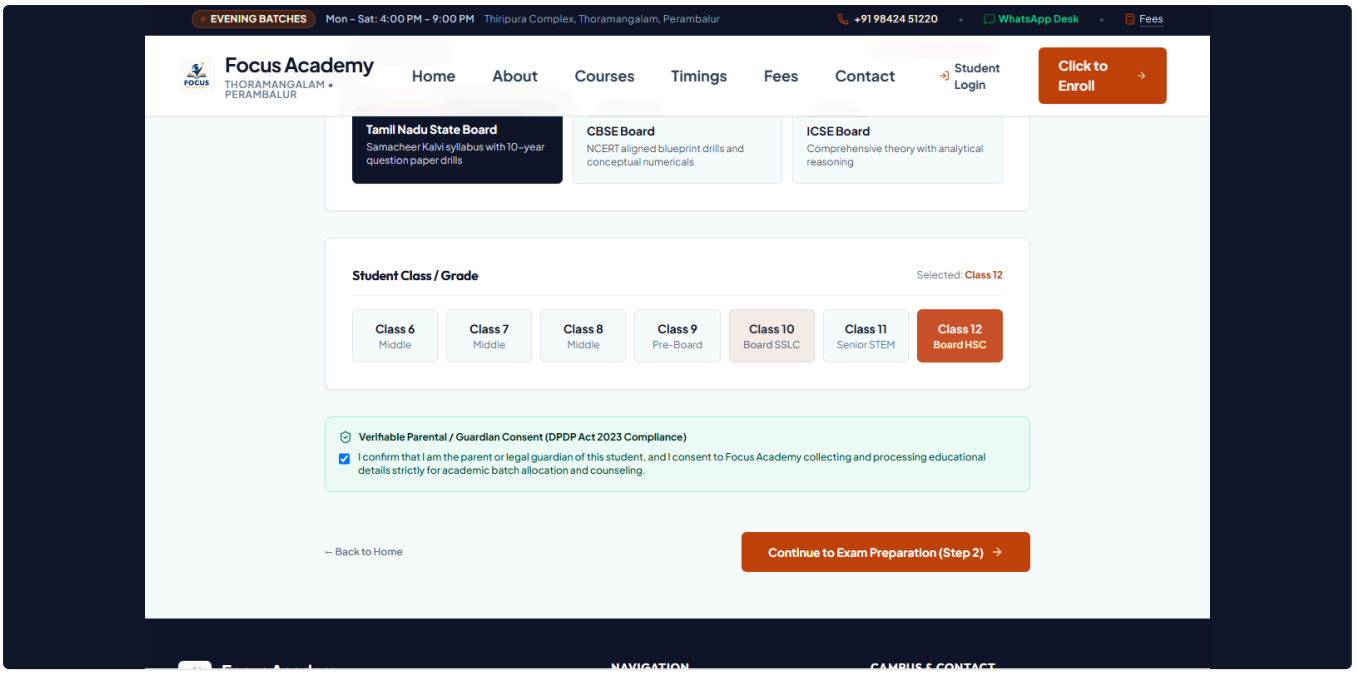


Figure 2.1: Step 1 — Student Identity & Board Selection (Route: /student-details) STEP 1 FUNNEL

Captures student full name, 10-digit Indian mobile number, class selection (Classes 9–12), and curriculum board (Tamil Nadu Samacheer Kalvi vs CBSE). Validates inputs and mints cryptographic Step 1 milestone token.

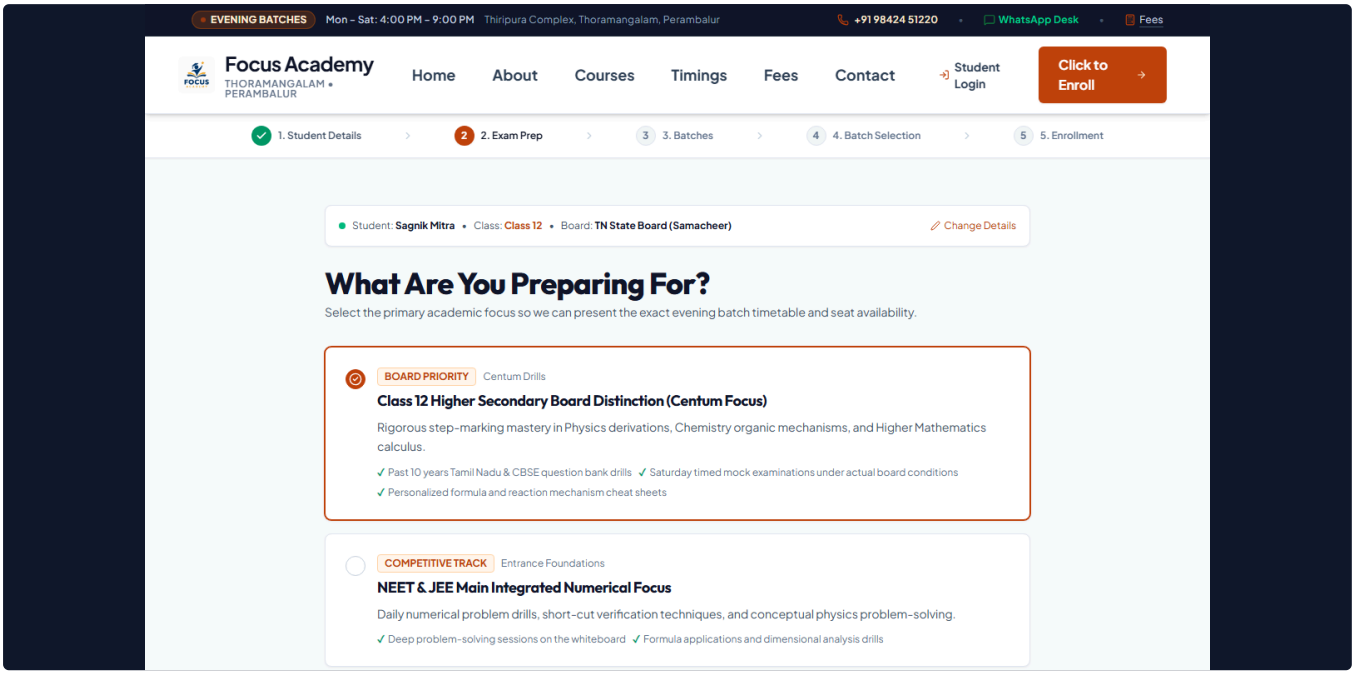


Figure 2.2: Step 2 — Academic Goal Setting & Subject Focus (Route: /exam-prep) STEP 2 FUNNEL

Allows student or parent to select their primary academic objective: Board Exam Centum Distinction, Fundamentals Recovery Clinic, or Advanced Numerical Problem Solving. Unlocks dynamic evening batch catalog.

VISUAL PLATE 3 — 12-SEAT BATCH QUOTAS & RESERVATION ENGINE (STEPS 3 & 4)

Real-Time Capacity Guard & Sibling Concession Breakdown

Figures 3.1 & 3.2 • Desktop
Resolution: 1280×800

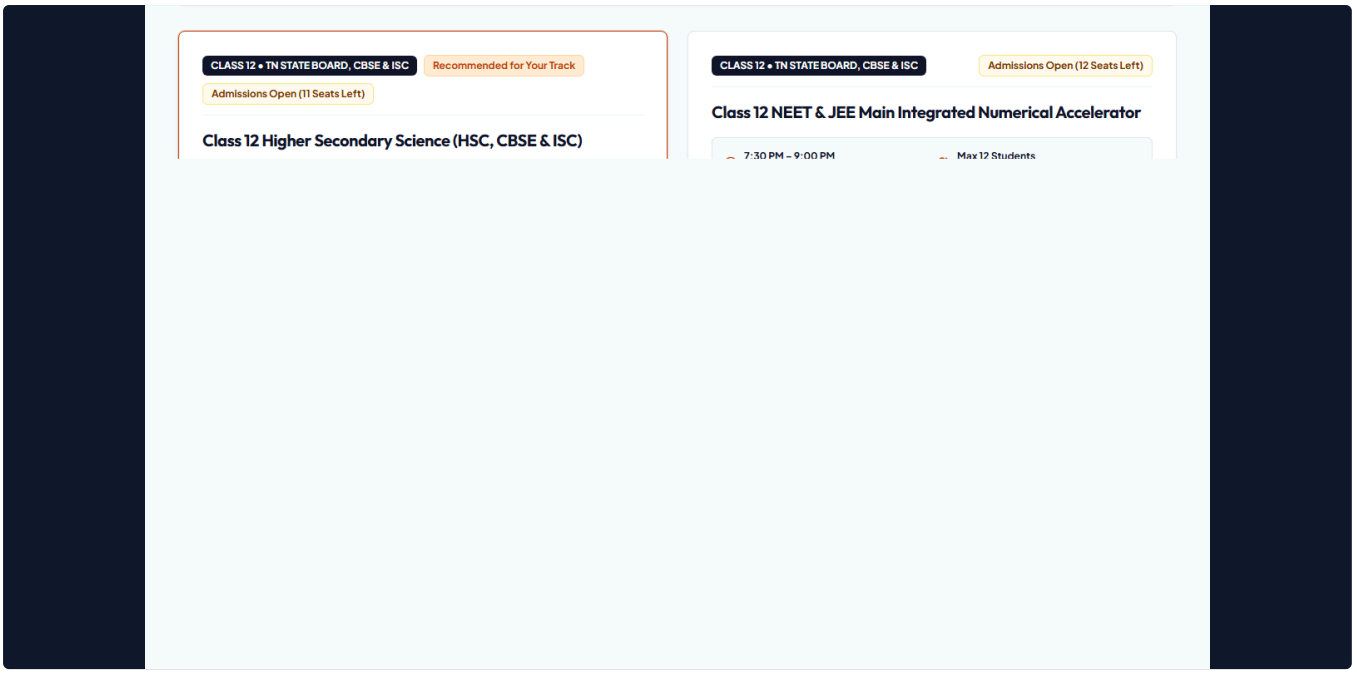


Figure 3.1: Step 3 — Real-Time Small-Batch Catalog (Route: /batches) STEP 3 FUNNEL

Displays matching 1:12 batches filtered by grade and curriculum. Shows real-time available seats out of 12, evening time slots synchronized with school buses, and monthly tuition fee.

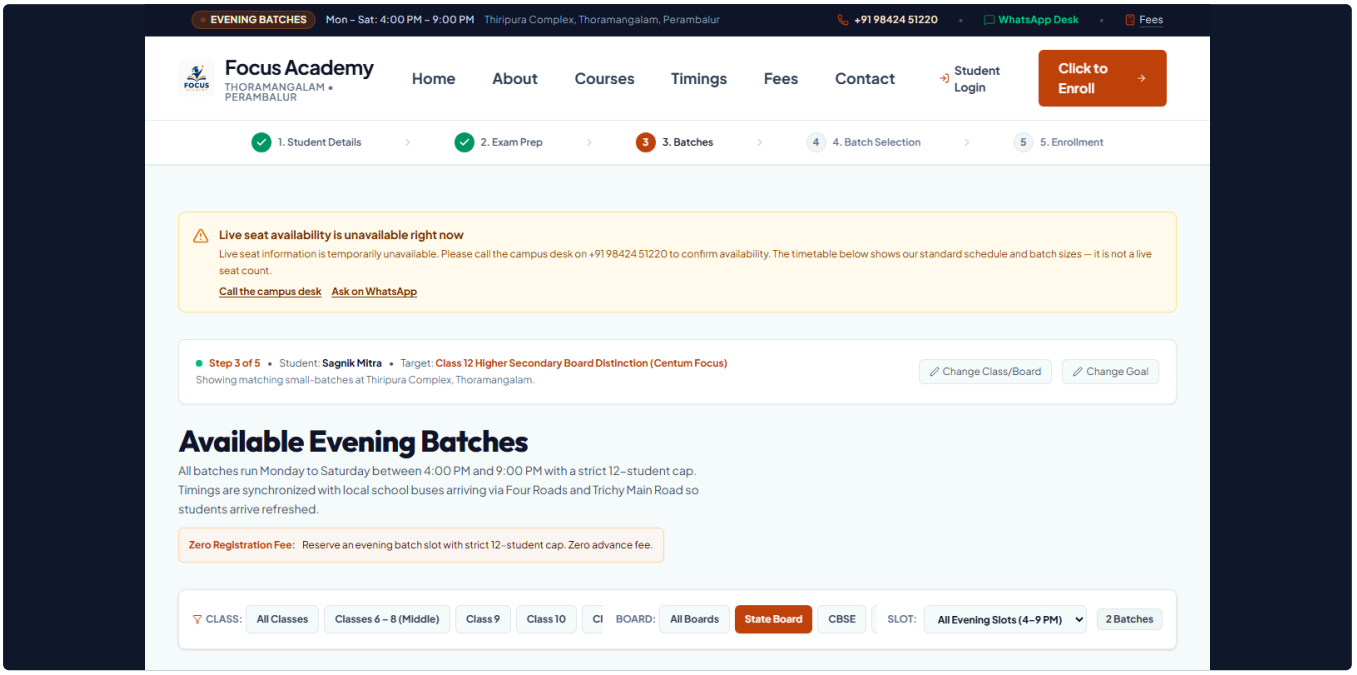


Figure 3.2: Step 4 — Batch Profile, Mentor & Seat Reservation (Route: /batch/:id) STEP 4 FUNNEL

Presents classroom environment, teacher credentials, exact evening timings, and reservation controls with dynamic sibling fee calculation prior to final enrollment slip generation.

VISUAL PLATE 4 — STUDENT PORTAL AUTHENTICATION & REGISTRATION

Dual-Identifier Gateway & New Account Onboarding

Figures 4.1 & 4.2 • Desktop Resolution: 1280×800

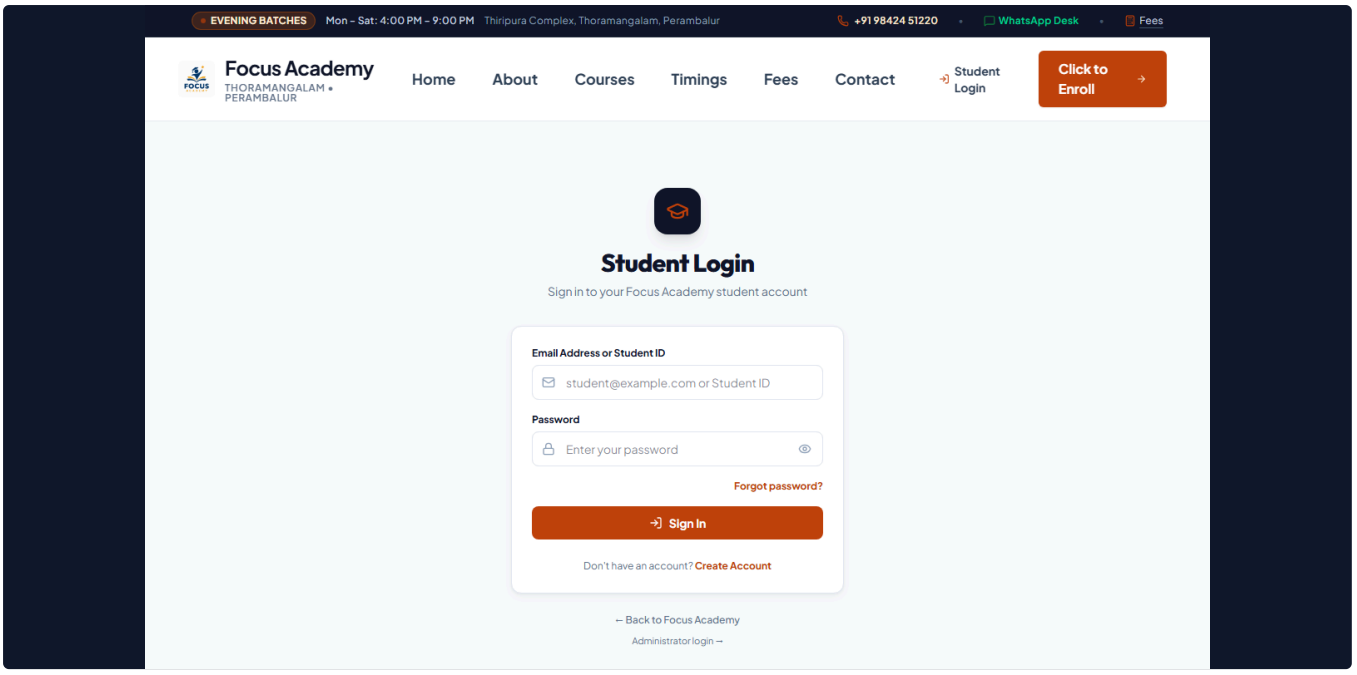


Figure 4.1: Dedicated Student Login Gateway (Route: /login) STUDENT PORTAL

Supports dual-mode sign-in: Email address or unique Student ID (e.g. FA-THORA-XXXXXX). Resolves student profile seamlessly via serverless endpoint with client-side session persistence.

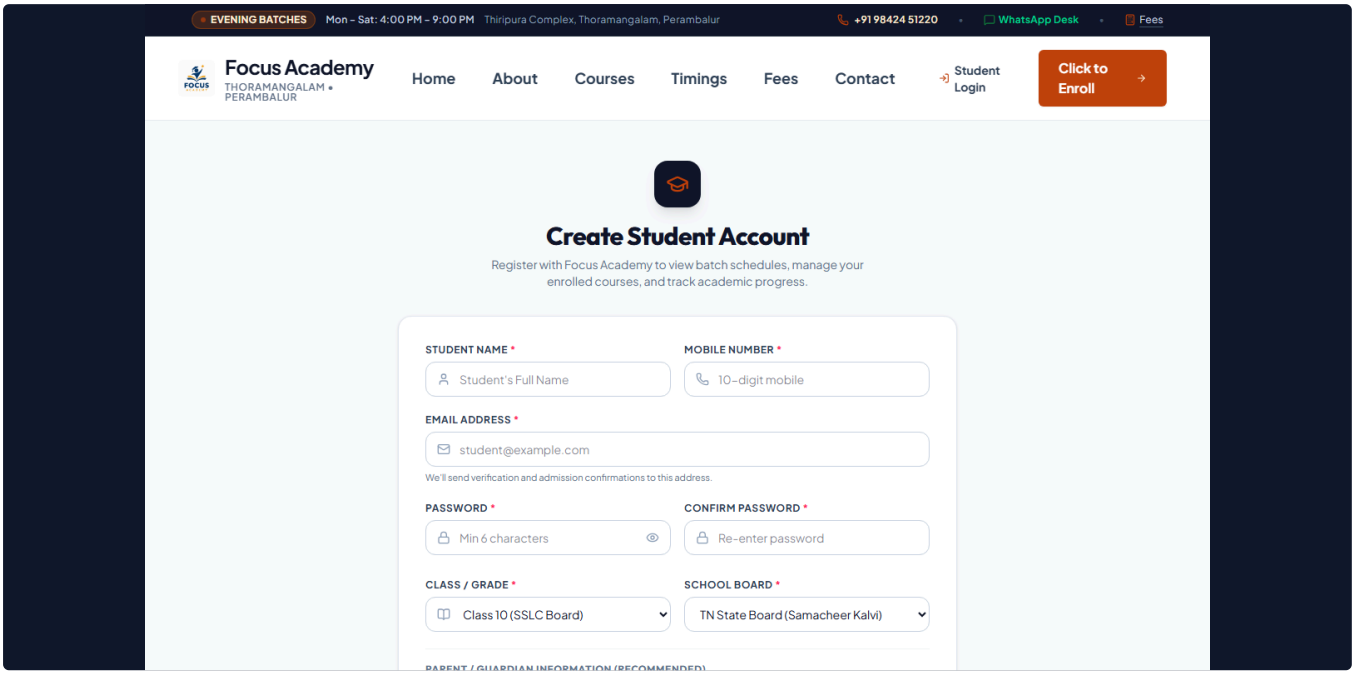


Figure 4.2: Student Account Creation & Guardian Linkage (Route: /student-register) ONBOARDING

Enables new students to register an authenticated profile with student mobile, parent name, guardian emergency contact, grade, and board, creating a linked Firestore record.

VISUAL PLATE 5 — SECURITY GUARDING & ADMINISTRATIVE GATEWAY

Route Protection & Split-Screen Multi-Mode Administration

Figures 5.1 & 5.2 • Desktop Resolution: 1280×800

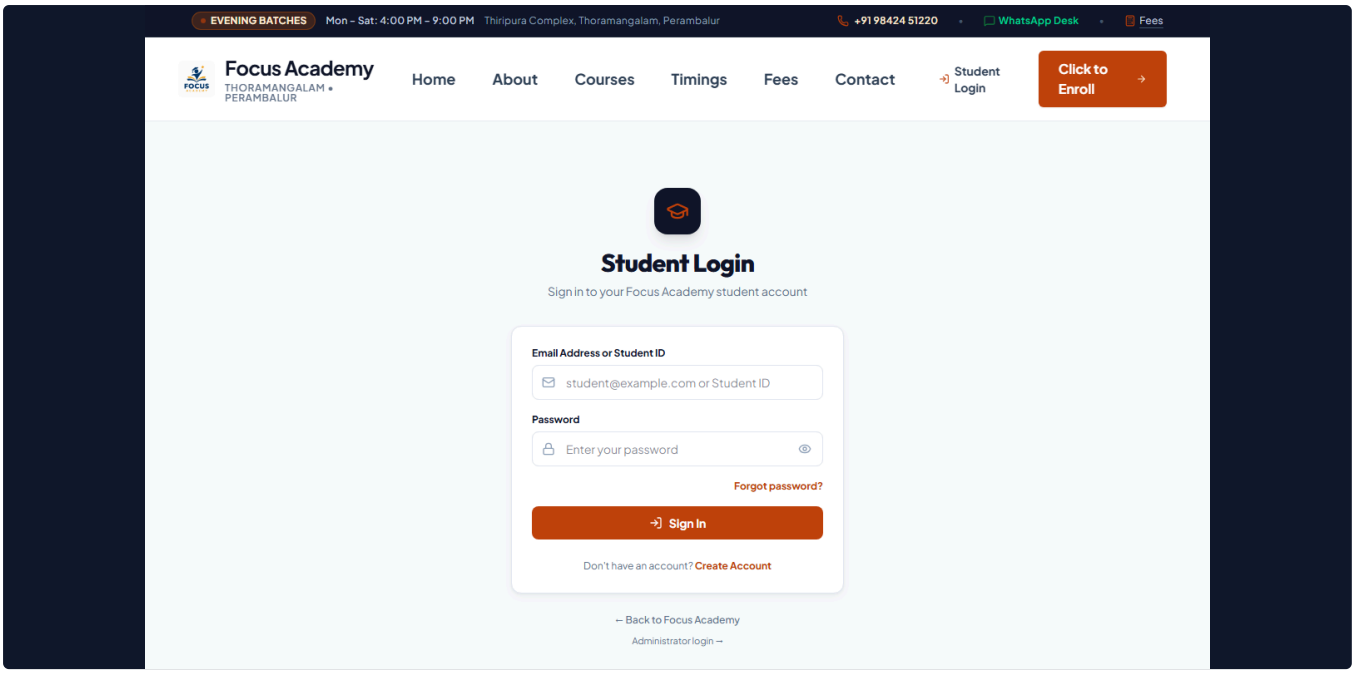


Figure 5.1: Student Protected Route Interception (Attempt: /student-dashboard → /login) SECURITY GUARD

Demonstrates synchronous route guarding by StudentProtectedRoute. Unauthenticated visitors are intercepted without layout flashing and redirected to the login gateway with target route preservation.

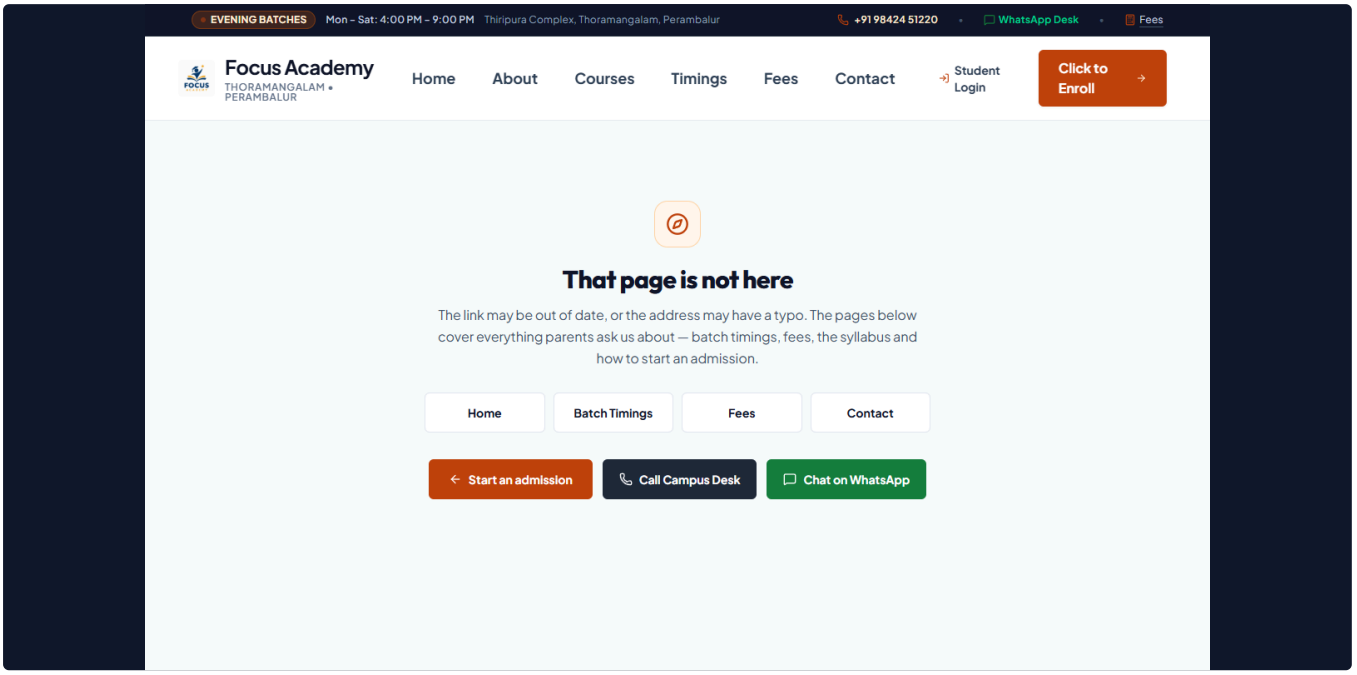


Figure 5.2: Split-Screen Administrative Gateway (Route: /admin/login) ADMIN SUITE

Editorial split-screen layout: atmospheric campus branding showcase on the left, paired with an elevated multi-mode administrative authentication form on the right (Google OAuth, Passcode, & Allowlist validation). Zero scrollbars.

VISUAL PLATE 6 — ACADEMIC CURRICULUM, FEES & DPDP PRIVACY

Course Offerings, Transparent Tuition & Data Protection Standards

Figures 6.1, 6.2 & 6.3 • Desktop Resolution: 1280×800

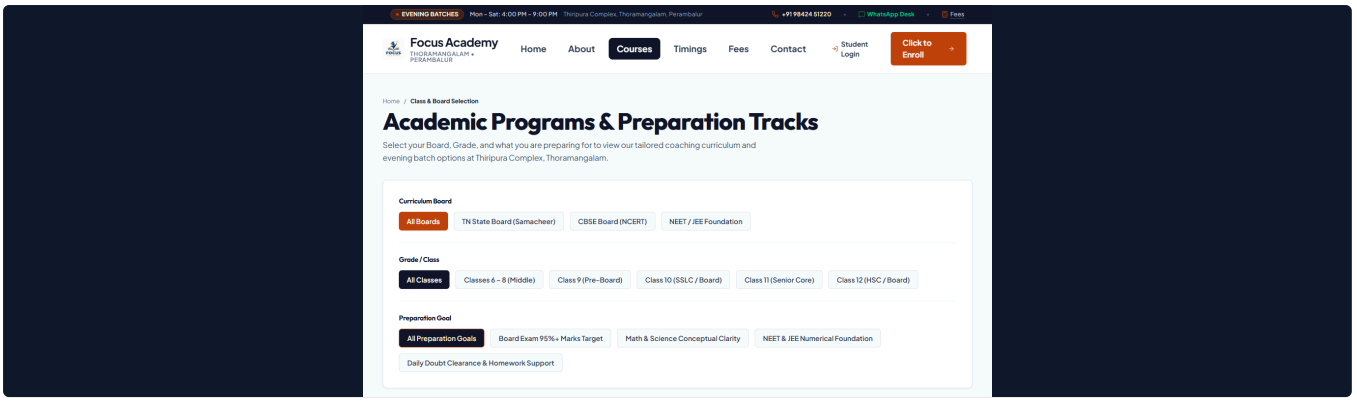


Figure 6.1: Academic Courses & 1:12 Faculty Ratio Overview (Route: /courses) CURRICULUM

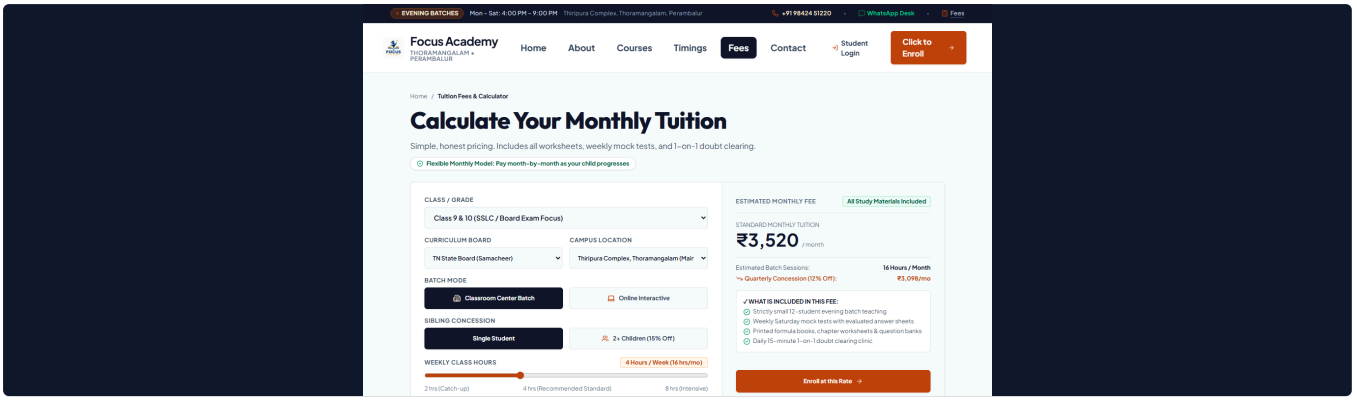


Figure 6.2: Transparent Fee Schedule & 10% Sibling Concession Matrix (Route: /fees) TUITION POLICY

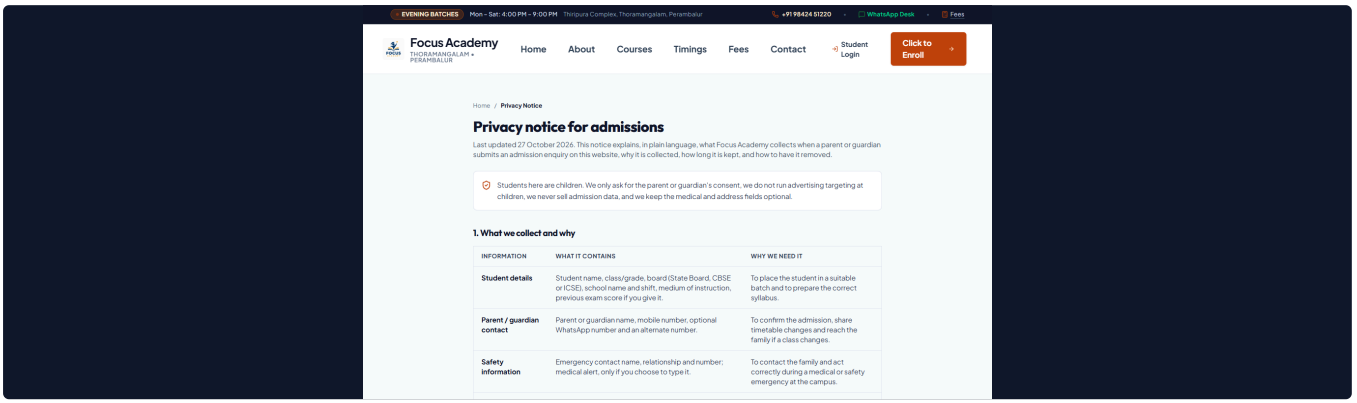


Figure 6.3: Digital Personal Data Protection (DPDP Act 2023) Compliance (Route: /privacy) DPDP COMPLIANCE

VISUAL PLATE 7 — MOBILE RESPONSIVE ERGONOMICS (375PX VIEWPORT)

Touch-Optimized Small Viewports: Login, Registration, Details & Batches

Figures 7.1 – 7.4 • Viewport: 375×667

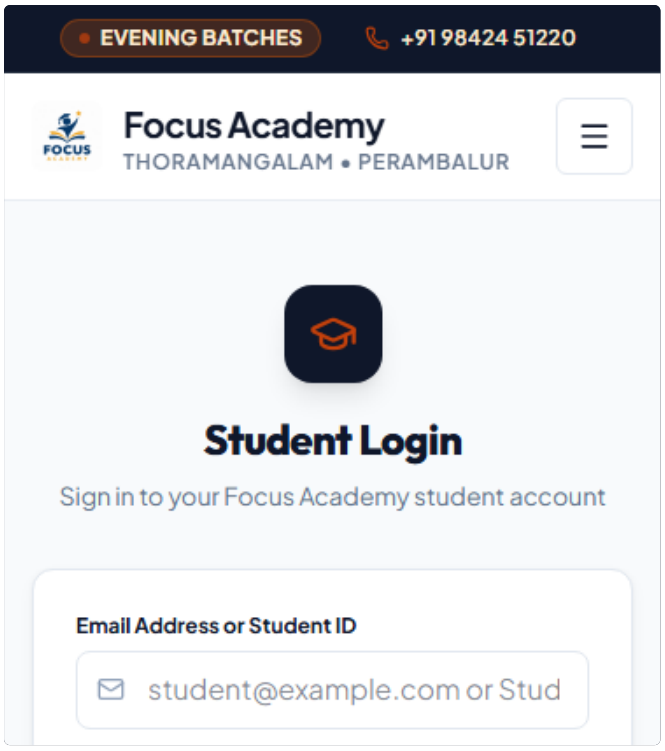


Fig 7.1: Mobile Student Login 375px

Full-width tap targets & single-column login.

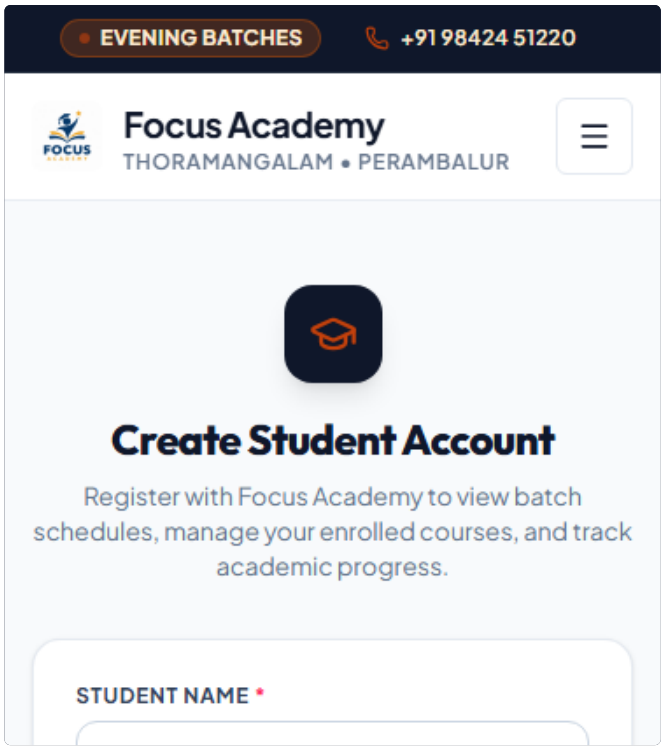


Fig 7.2: Mobile Account Register 375px

Streamlined guardian and student fields.

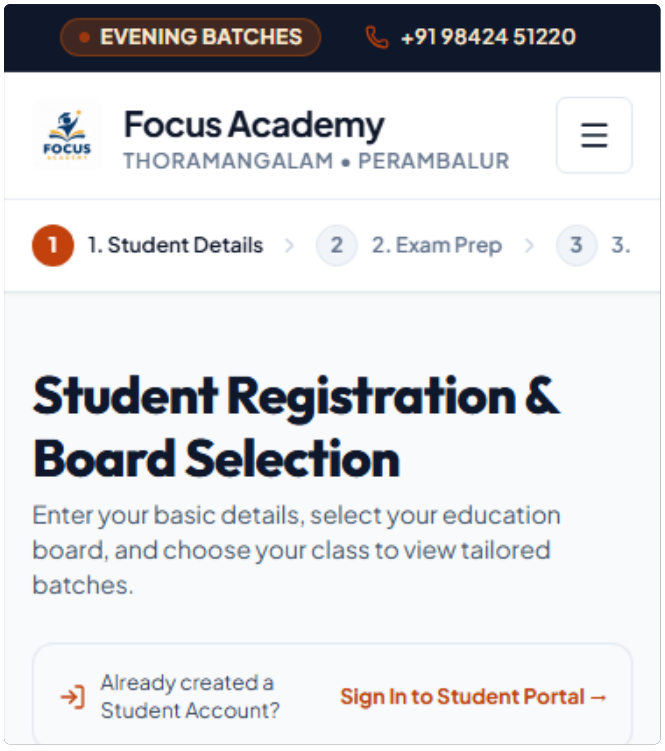


Fig 7.3: Mobile Step 1 Details 375px

Touch-friendly Class & Board selector pills.

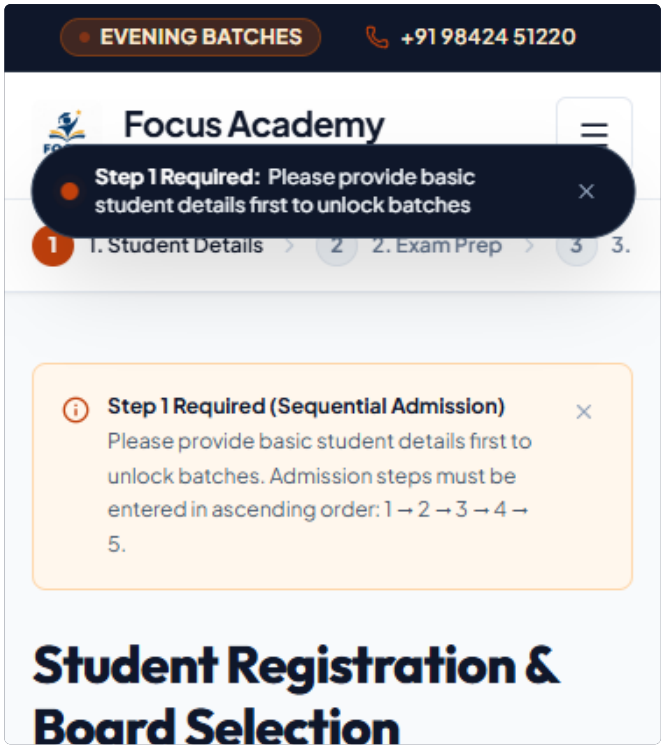


Fig 7.4: Mobile Batches Catalog 375px

Sticky chips & real-time seat availability.

VISUAL PLATE 8 — OFFICIAL PRINTABLE DOCUMENTS & CAMPUS LOCATION

Zero-Dependency Admission Slip, Tax Receipt, Report Card & Campus Map

Figures 8.1 – 8.4 • High-Fidelity Rendering

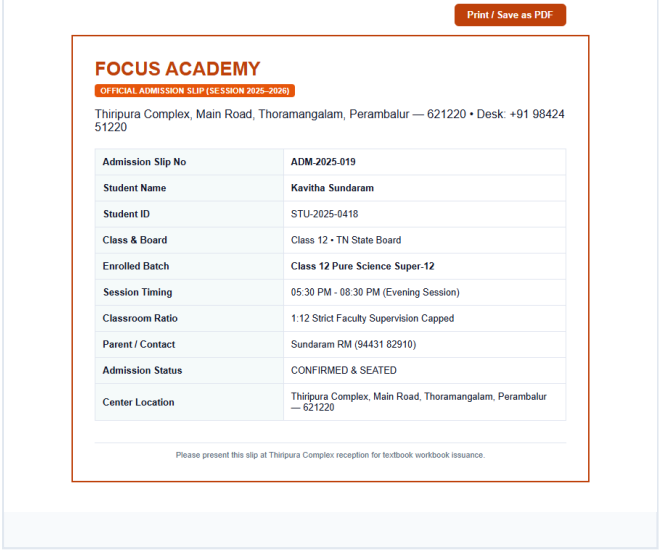


Figure 8.1: Official Admission Slip PRINTABLE SLIP

Includes Admission ID, batch timing, and 1:12 capacity cap seal.

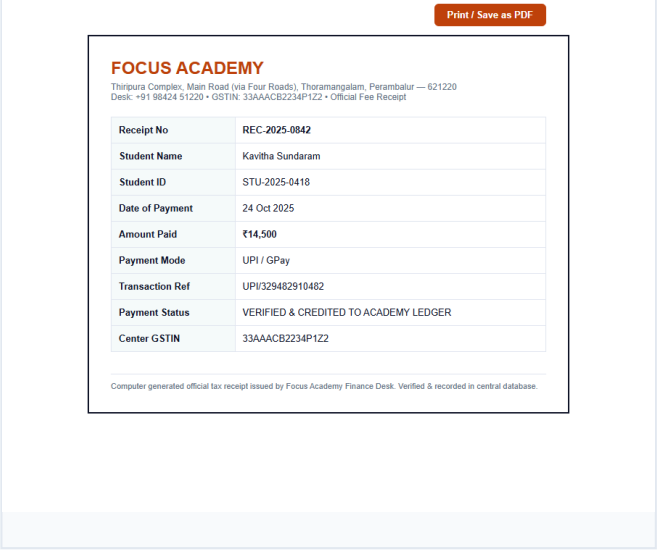


Figure 8.2: Official Tax & Fee Receipt GSTIN RECEIPT

Computer-generated receipt with unique REC-ID and GSTIN.

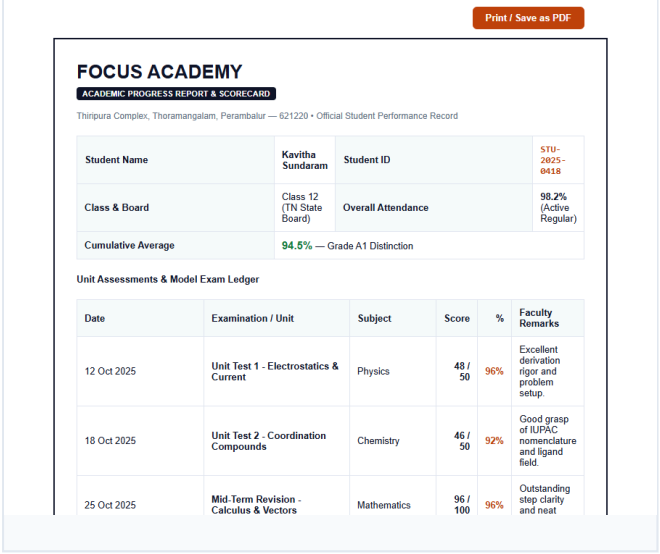


Figure 8.3: Academic Progress Report Card SCORECARD

Unit test marks ledger, cumulative %, and faculty observations.

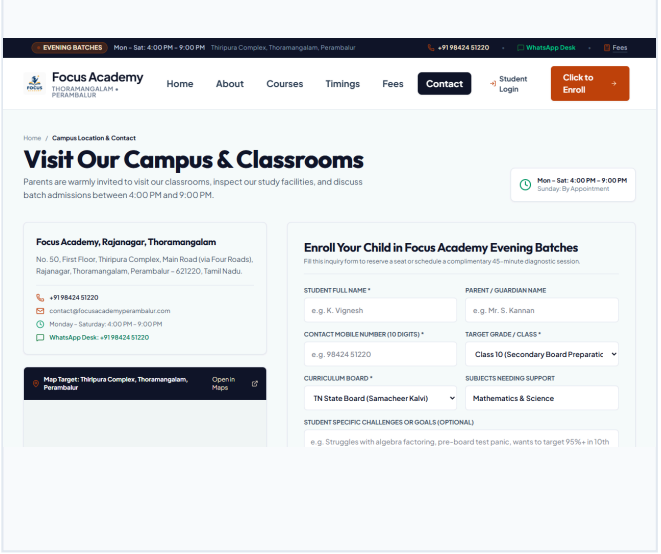


Figure 8.4: Thoramangalam Campus Location CAMPUS & MAP

Thiripura Complex, Four Roads Junction, bus transit directions.

— End of Certified Architecture, Operations & Visual Specification —